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NeuroDISSCONNECT
(No pun intended)
presents:

Spectro-Temporal Encoding of Continuous Kinematics in Human Motor Cortex



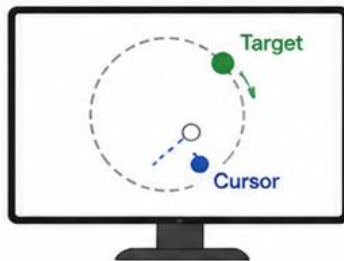
NeuroCONNECT
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Introduction



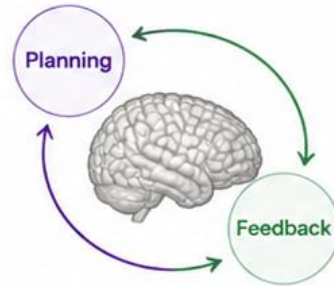
Natural movement is **not discrete**.
It is a **dynamic process** that changes **rapidly** over time.

1. Natural movement is continuous



- Movement evolves **smoothly** over time.
- The state is **never static**.

2. The brain continuously plans and corrects



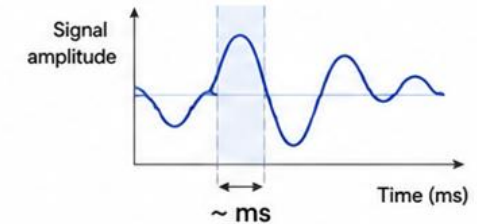
- The brain needs **planning** and **feedback** to correct motion.

3. Rich information is constantly monitored



- This information includes **position**, **direction**, **velocity**, **acceleration**, and **error**.

4. These changes are very rapid



- Behavioral and neural changes occur on the **millisecond scale**.



Thus, we need a measurement that captures these rapid changes.

ECoG does this for us.



Literature Review & Theoretical Context



Schalk et al. (2007)

- ECoG can decode continuous 2D cursor trajectories.
- LMP is associated with position and velocity.



Hammer et al. (2013)

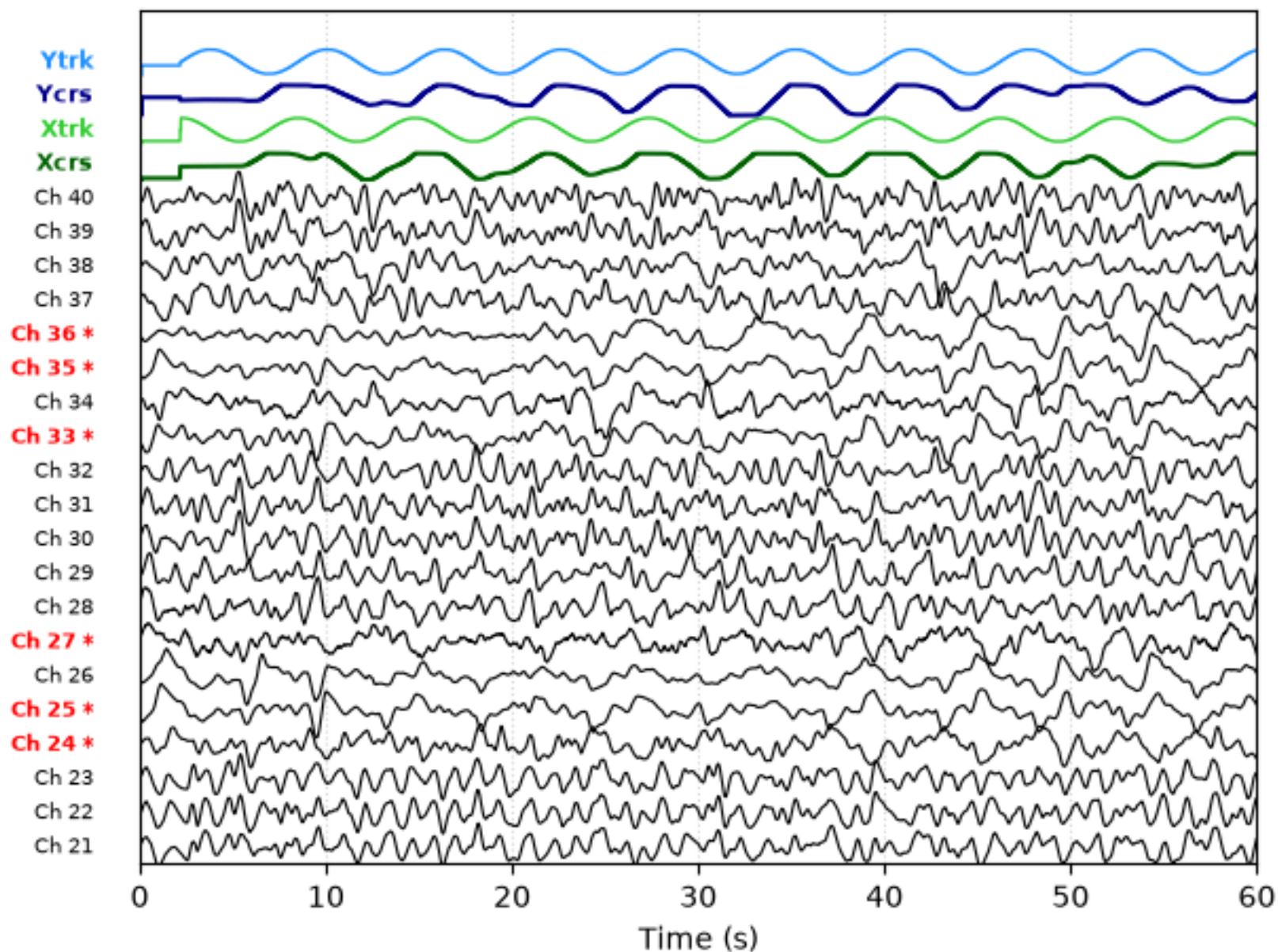
- Position and velocity are better represented than acceleration.

THE RESEARCH GAP

Prior decoding models evaluate position/velocity in isolation, or focus solely on high-gamma power.

However, how different kinematic features are distributed across cortical frequency bands and temporal scales remains unclear.

A) ECoG Time Course (Channels 21-40) & Kinematics



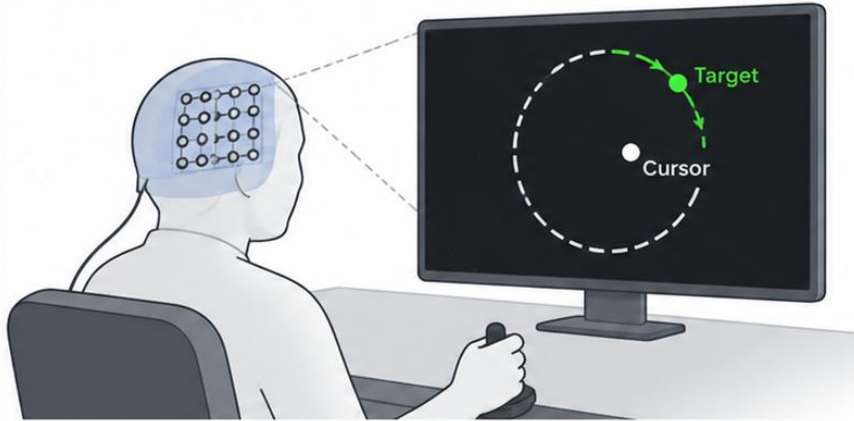
Research Questions

#	Dimension	Forward Encoding Frame: X kinematics → Y neural
1	Spectral Specialization	Do low-frequency potentials (LMP < 4 Hz) and high-frequency power envelopes encode different aspects of movement dynamics?
2	Temporal Lag Profiles	What are the neural lead–lag relationships between spectral features and kinematic variables?
3	Encoding Stability	How stable is spectro-temporal kinematic encoding across continuous tracking sessions?

Task Description

Continuous 2D Joystick Tracking

Participants control a cursor with a hand-held joystick to continuously track a moving target on the screen.



4 Subjects

(A, B, C, D) from the original study dataset



Subdural ECoG

48–64 electrodes (motor-related cortex)



Simultaneous recording

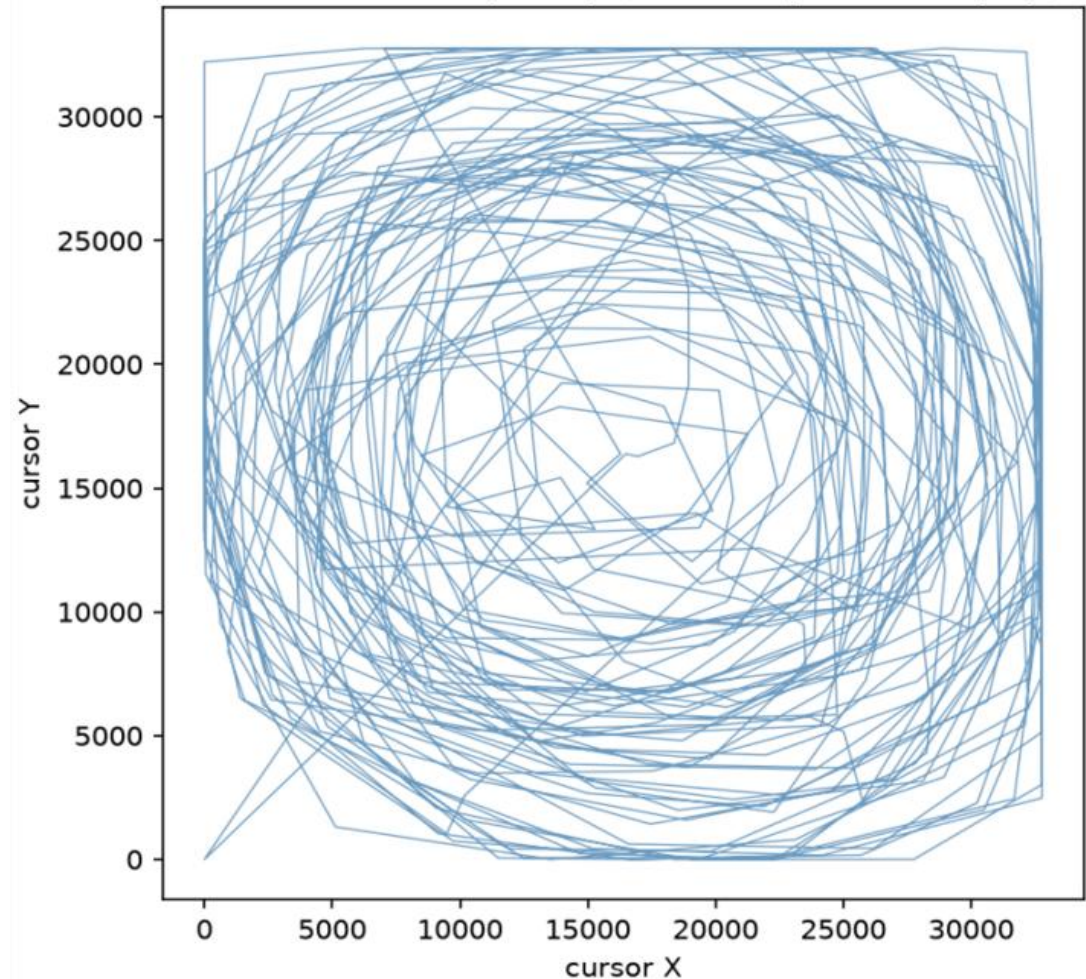
ECoG + cursor & target signals



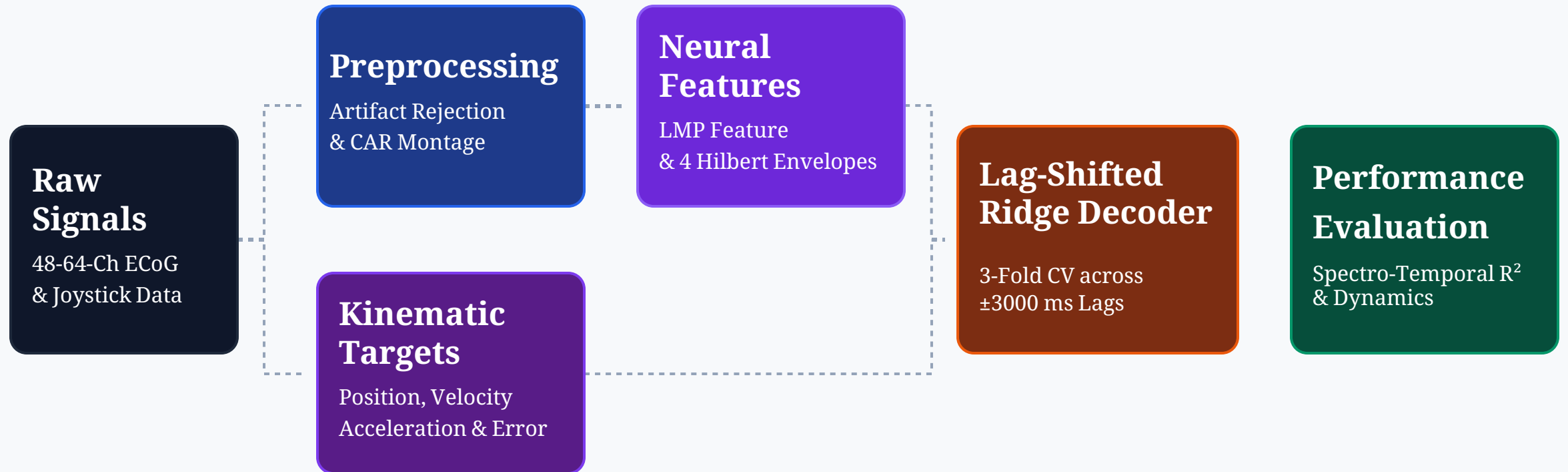
Sampling rate

1000 Hz

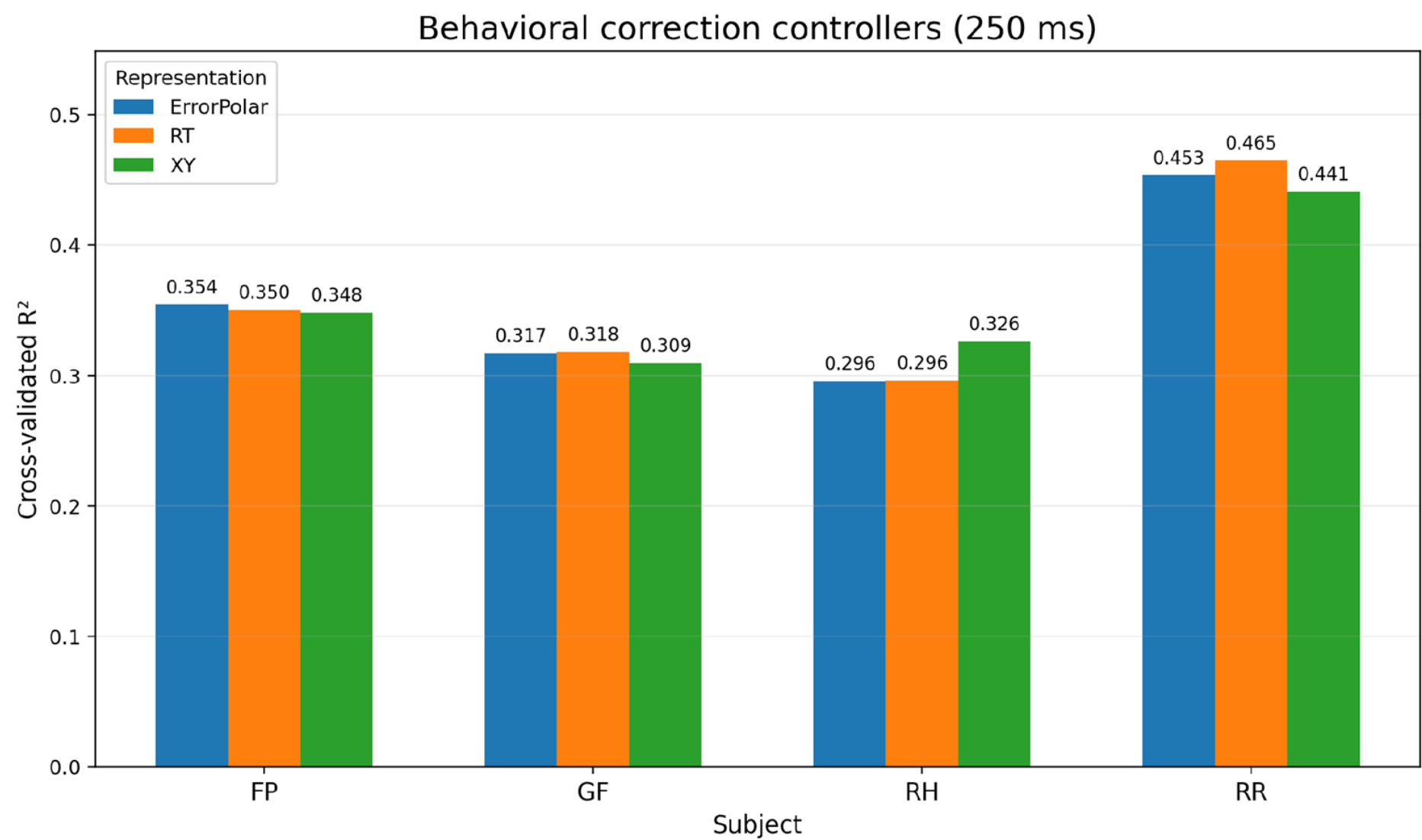
Block 0: cursor trajectory (downsampled for display)

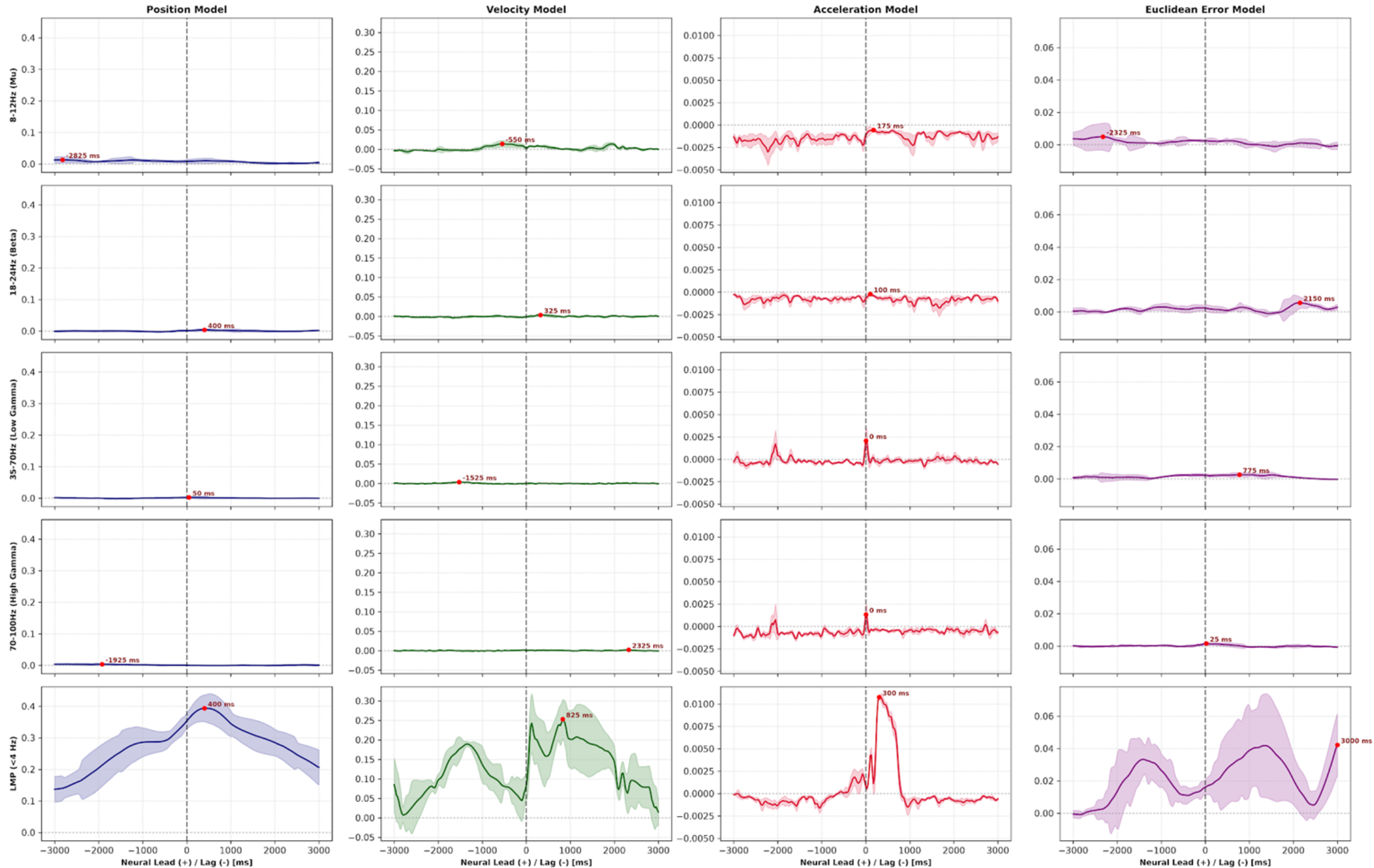


Method - Pipeline



Bonus Idea

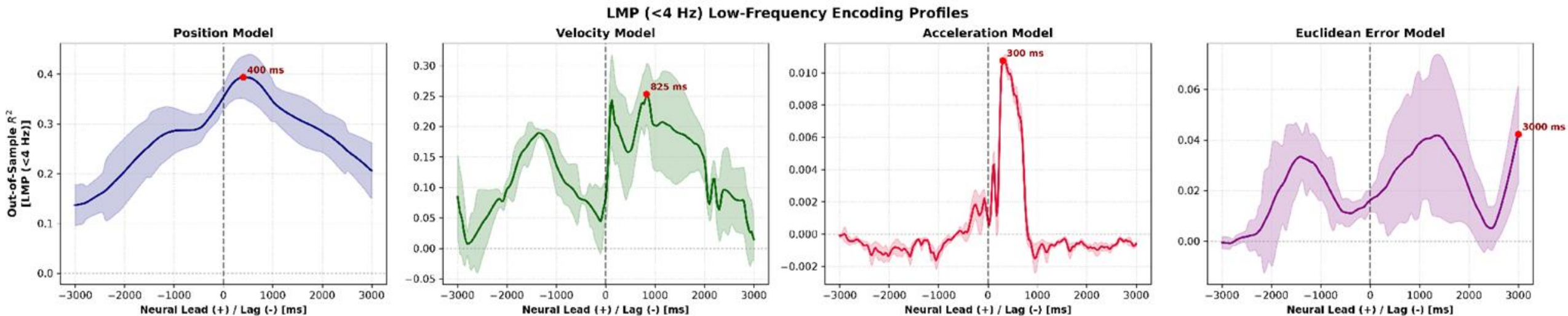




Result(1/3) —Low-Frequency Dominance (LMP Encoding)

Finding 1: LMP Provides Sustained Information About Movement Trajectory

Low-frequency activity (LMP) showed the strongest decoding performance across continuous kinematic variables.



Interpretation



Low-frequency cortical activity captures slowly varying movement dynamics. The broad temporal profile suggests that LMP contains information related to:

- Continuous trajectory evolution and movement dynamics.
- Movement preparation and execution dynamics.
- This pattern differs from transient motor responses represented by higher-frequency activity.

Key References

Hammer et al. (2013)

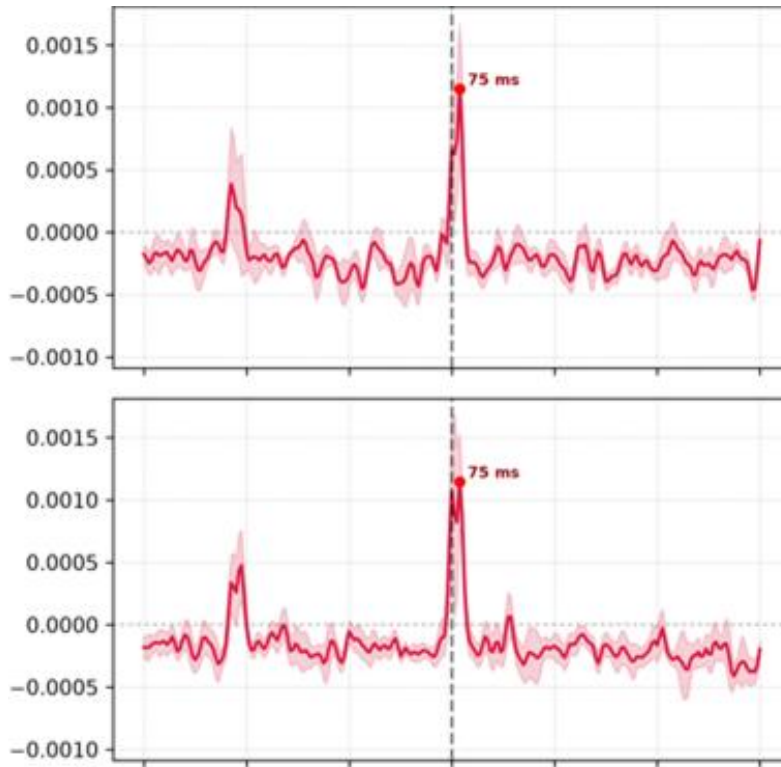
Flint et al. (2010)



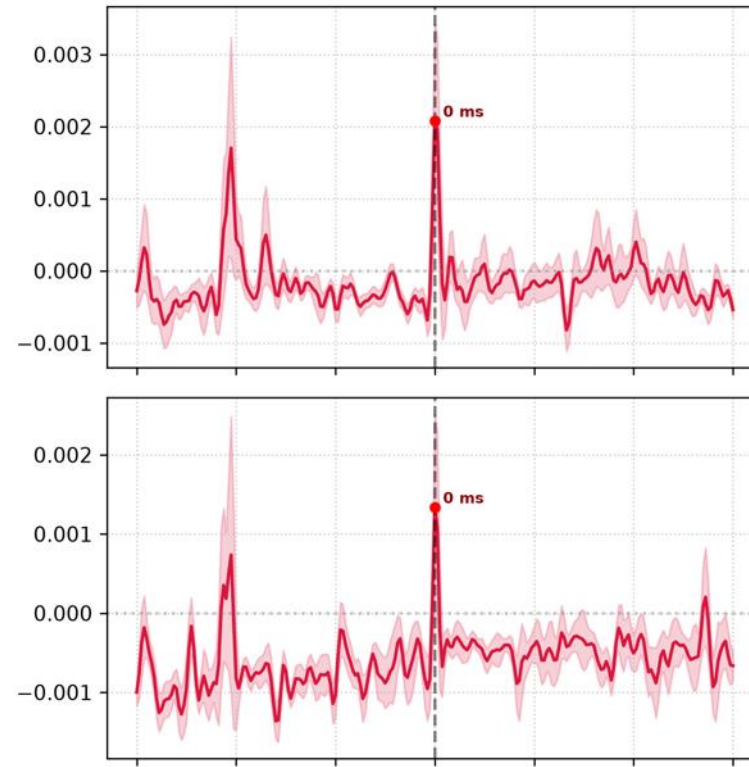
Result(2/3) —Gamma Temporal Precision for Rapid Movement Changes

Finding 2: Gamma Activity Shows Temporally Localized Information About Acceleration
High-frequency activity showed a distinct temporal decoding pattern compared with LMP.

Subject 2



Subject 3



Acceleration Model in Low (row 1) and High (row 2) Gamma Band

insight Interpretation

Gamma activity provides **temporally precise information** associated with rapid changes in movement state.

High-gamma power directly **mirrors local multi-unit population spiking** in primary motor cortex (M1).

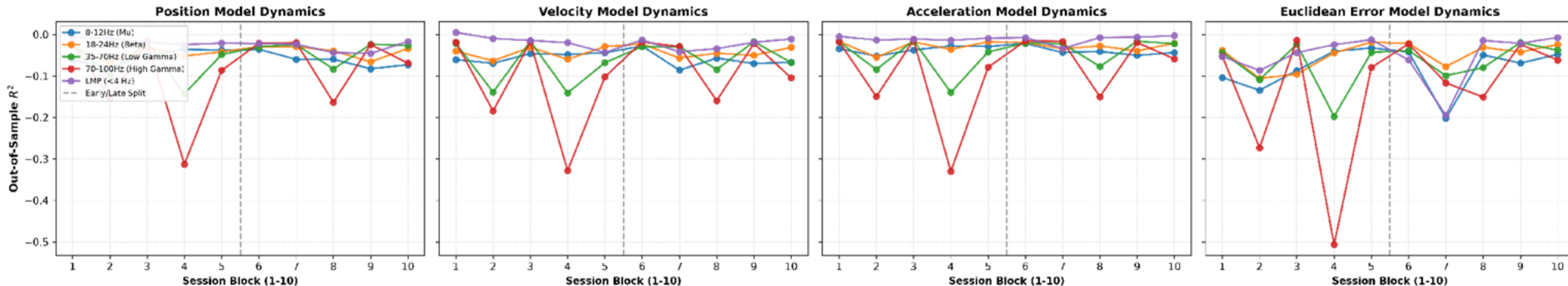
Key Reference

Wolpet et al. (2000)

Result(3/3) — Within-Session Non-Stationarity of Neural Decoding

Finding 3: Neural–Behavioral Mapping Changes Across the Recording Session
High-frequency activity showed a distinct temporal decoding pattern compared with LMP.

Temporal Adaptation & Encoding Stability Across 10 Session Blocks



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Interpretation

- Changes in neural population dynamics
- Variations in attention or engagement
- Changes in motor strategy

Key References

Perge et al. (2013)

Degenhart et al. (2020)



Takeaway Message



Spectro-Temporal Organization of Motor Information

Motor information is distributed across distinct temporal scales: LMP provides sustained trajectory-related information, while gamma activity shows brief, time-precise responses to rapid motor changes.



Future Directions

Implement Recurrent Neural Networks (RNNs, LSTMs) or Temporal Convolutional Networks (TCNs) to capture complex, non-linear spatiotemporal dynamics.

Expand target parameters to multi-joint 3D kinematics, grip force, and tactile feedback integration.



**THANK YOU!
FOR YOUR TIME...**

Hold your
applause,
I know.

**ANY
QUESTIONS?**

Please
say no...

